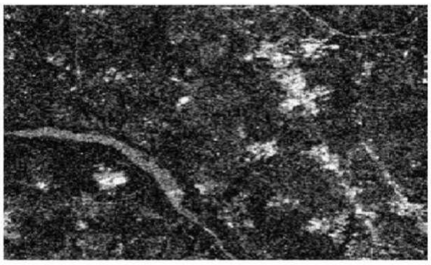
ITA05-ACTIVITY

SCENARIO BASED ACTIVITY

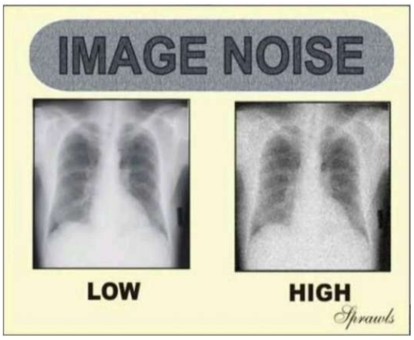
Mention the techniques used with reason:

1. A CCTV image contains **salt-and-pepper noise** due to transmission errors.

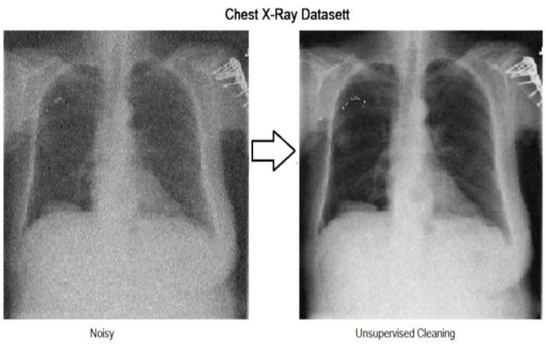
2. A satellite image is corrupted by **Gaussian noise** from sensors.



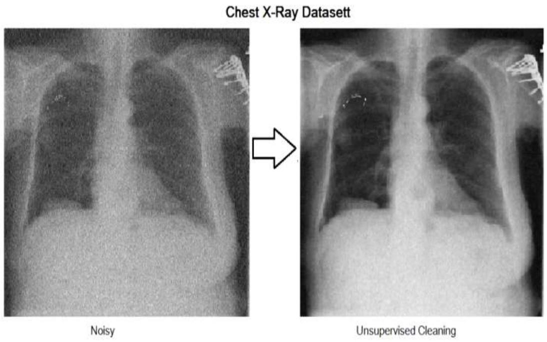
3. A medical X-ray image has random noise but fine edges must be preserved.



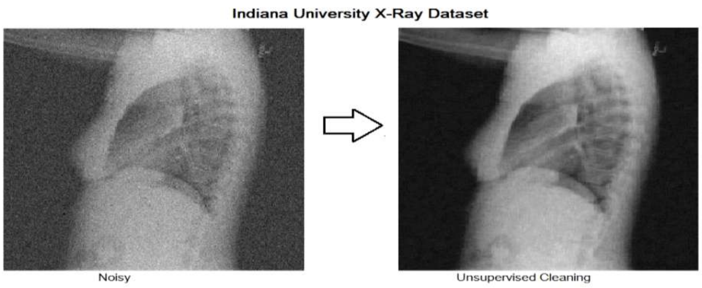
4. A scanned document has **black and white dots** scattered randomly.



4. A scanned document has **black and white dots** scattered randomly.



5. A low-light image shows heavy grain without clear boundaries.



6. A foggy road image has **very low contrast**.

7. An underwater image appears dull and washed out.

8. A chest X-ray shows poor visibility of internal structures.

6. A foggy road image has very low contrast.

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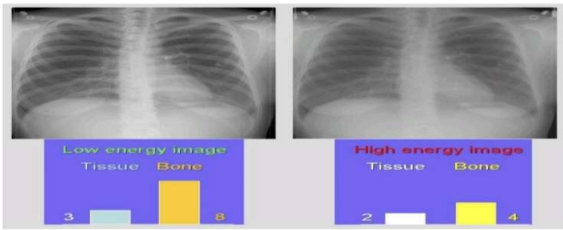
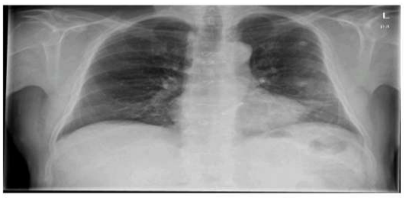


9. A night-time traffic image lacks intensity variation.



10. A grayscale image uses only a **narrow intensity range**.

30/07/26

① A CCTV image contains salt-and-pepper noise due to transmission errors.

Technique used :- Median Filter

Reason :- The median filter effectively removes salt-and-pepper noise by replacing each noisy pixel with the median value, while preserving image edges.

② A satellite image is corrupted by Gaussian noise from sensors.

Technique used : Gaussian Filter.

Reason : Smooths Gaussian noise effectively and reduces random sensor noise.

③ A medical x-ray image has random noise but fine edges must be preserved.

Technique used : Bilateral Filter.

Reason : Removes noise while preserving important edges.

④ A scanned document has black and white dots scattered randomly.

Technique used : Median filter

Reason : Eliminates black and white dot noise without affecting text.

⑤ A low-light image shows heavy grain without clear boundaries.

Technique used : Adaptive Median filter

Reason : Reduces heavy grain and preserves image details better.

⑥ A foggy road image has very low contrast

Technique used : Histogram Equalization

Reason : Enhances contrast and improves visibility.

⑦ An under water image appears dull and washed out

Technique used : Contrast stretching

Reason : Increases image contrast and improves brightness.

⑧ A chest x-ray shows poor visibility of internal structures.

Technique used : CLAHE [Contrast Limited Adaptive Histogram Equalization]

Reason : Enhances local contrast and reveals internal structures clearly.

⑨ A night-time traffic image lacks intensity variation.

Technique used : Histogram Equalization

Reason : Expands local contrast and reveals internal structures.

⑩ A gray scale image uses only a narrow intensity range.

Technique used : Contrast stretching

Reason : Expands the narrow intensity range to improve overa[ll] image contrast.